

Bioinformatics for engineers

Introduction to genetic distances

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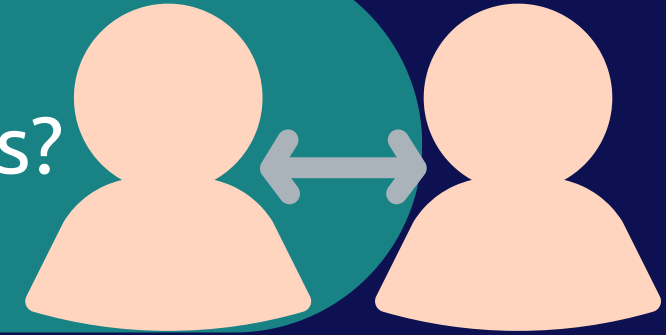


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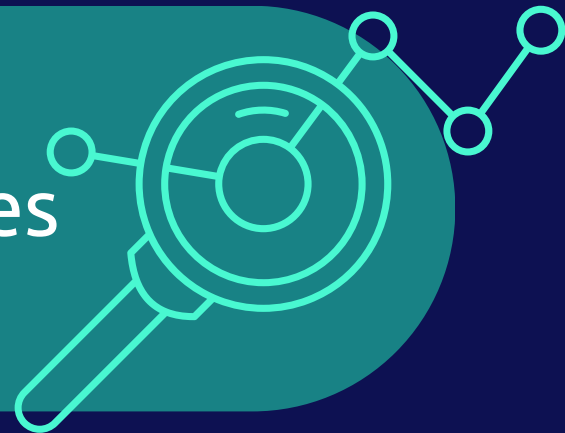
1. What is distance when talking about genes?



2. Why do we measure distances?



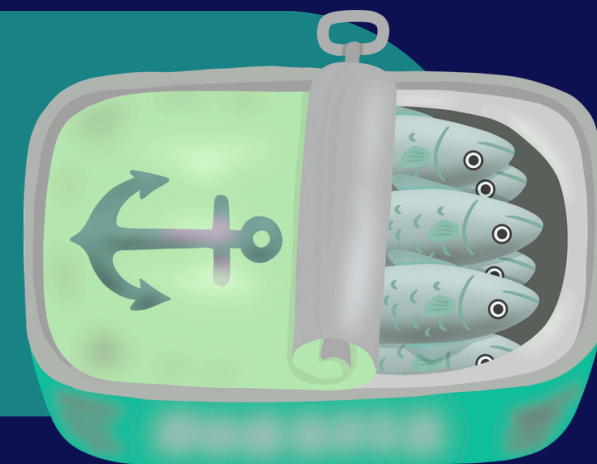
3. Metrics for genetic distances



4. The main outcome: Trees



5. Clustering



GENETIC DISTANCE

What is distance when we have genes?

In genomics and bioinformatics, there are two main definitions of what genetic distance means.

A measure of the evolutionary divergence between **homologous genes** with a common ancestor

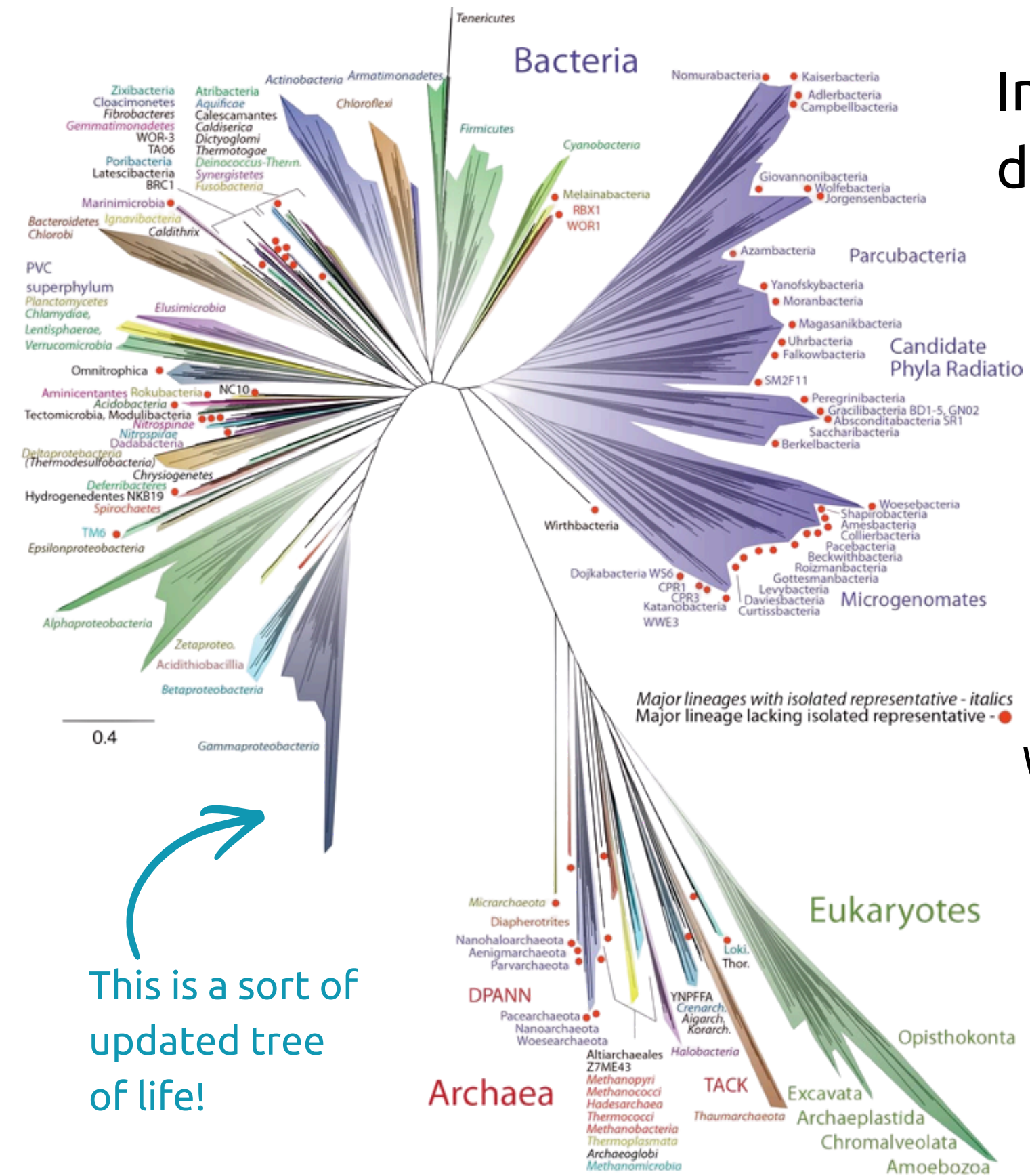
A measure of the degree of genetic differentiation between **populations**

We can reconcile these two valid ideas with this definition

"The extent of gene differences between populations or species that is measured by some numerical quantity" (Nei, 1987)

*Essentially any **quantitative measure of genetic difference**, be it at the sequence level or the allele frequency level, that is calculated between individuals, populations or species, can be defined as a genetic distance.*

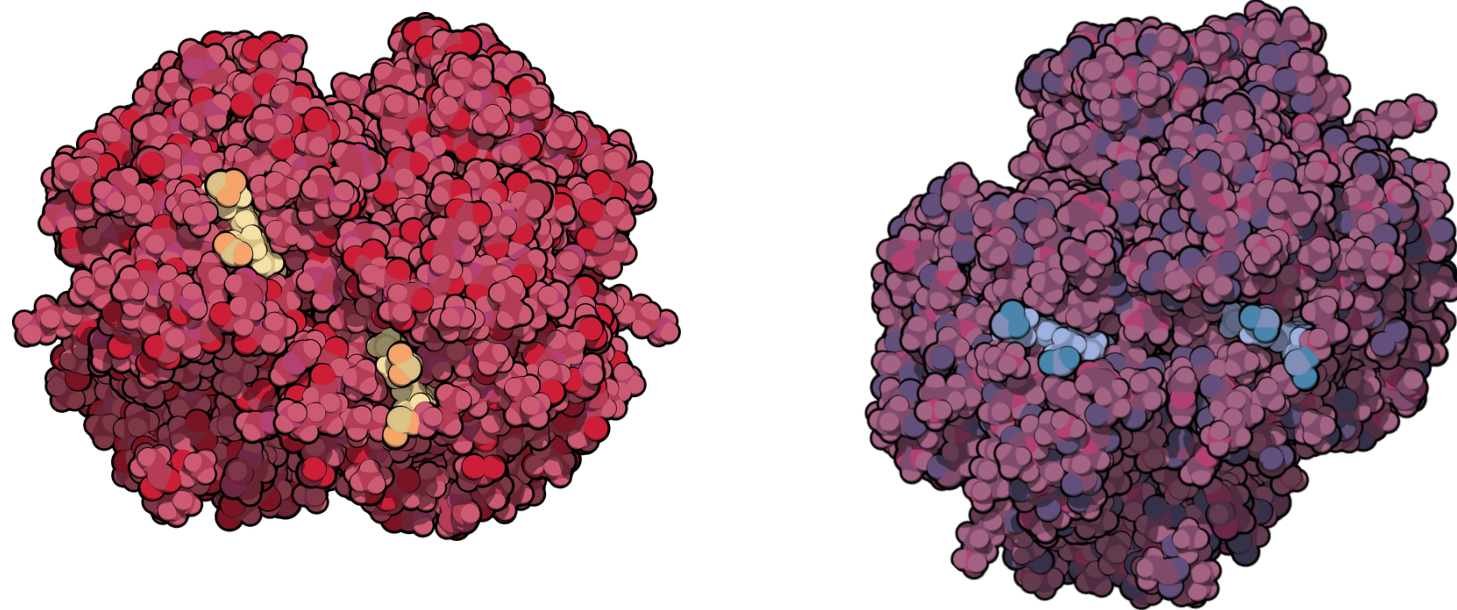
This is a sort of updated tree of life!



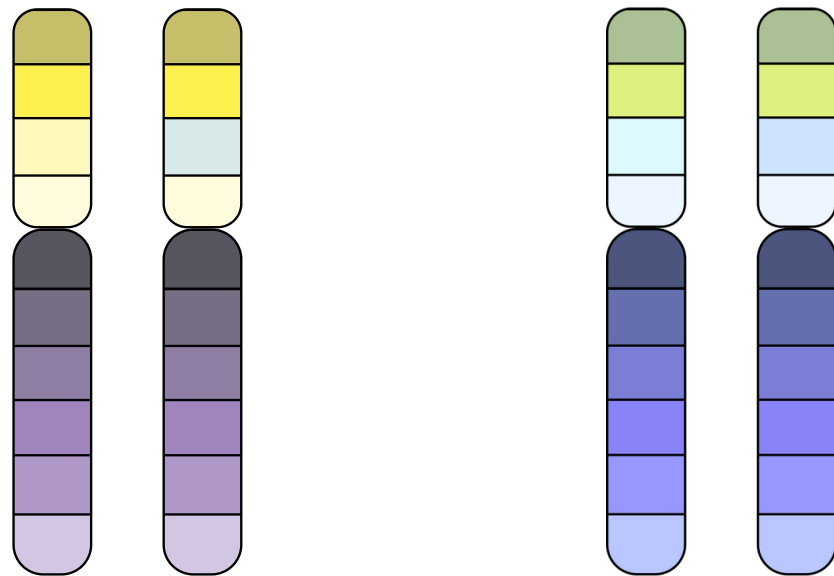
What is being compared?



Nucleotides in aligned sequences.

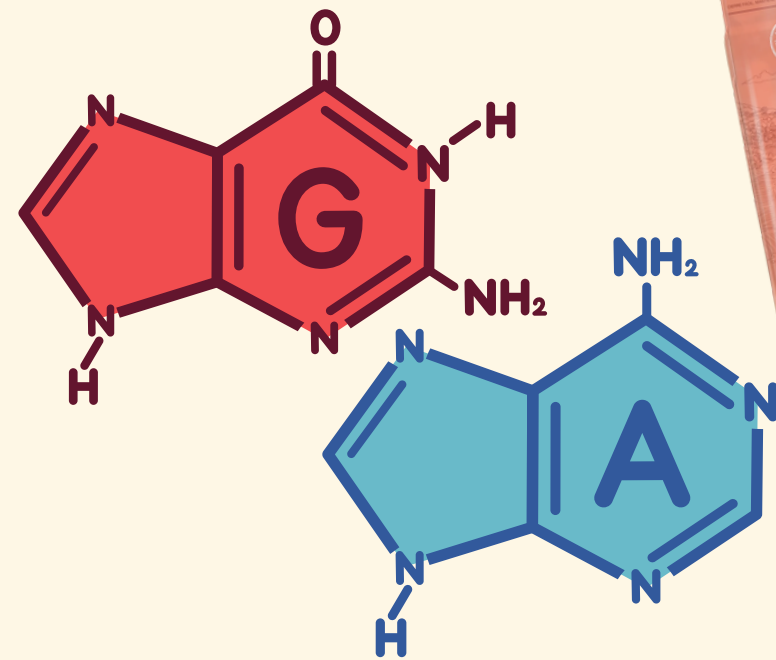


Aminoacids in homologous proteins.

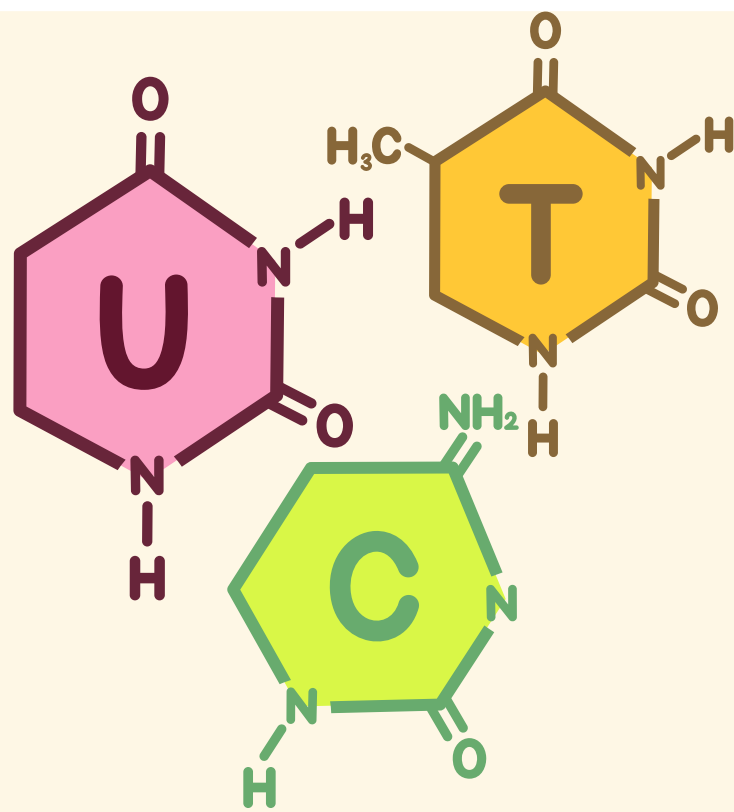


Allele frequencies among populations.

Focusing on nucleotides...

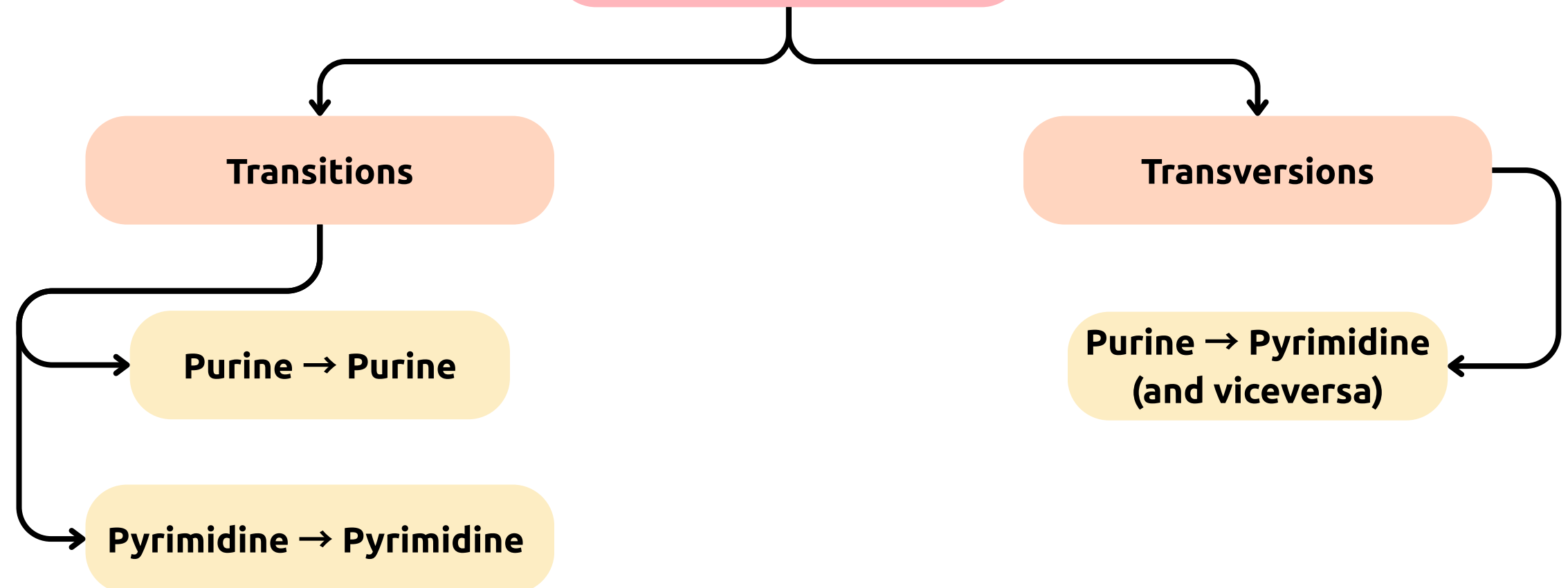


Purines



Pyrimidines

Nucleotide substitutions



The number of observed substitutions underestimates the actual substitutions due to multiple, parallel, and reverse substitutions → hence the need for correction models.

**BUT WHY DO WE MEASURE
GENETIC DISTANCES?**

Time is not observable

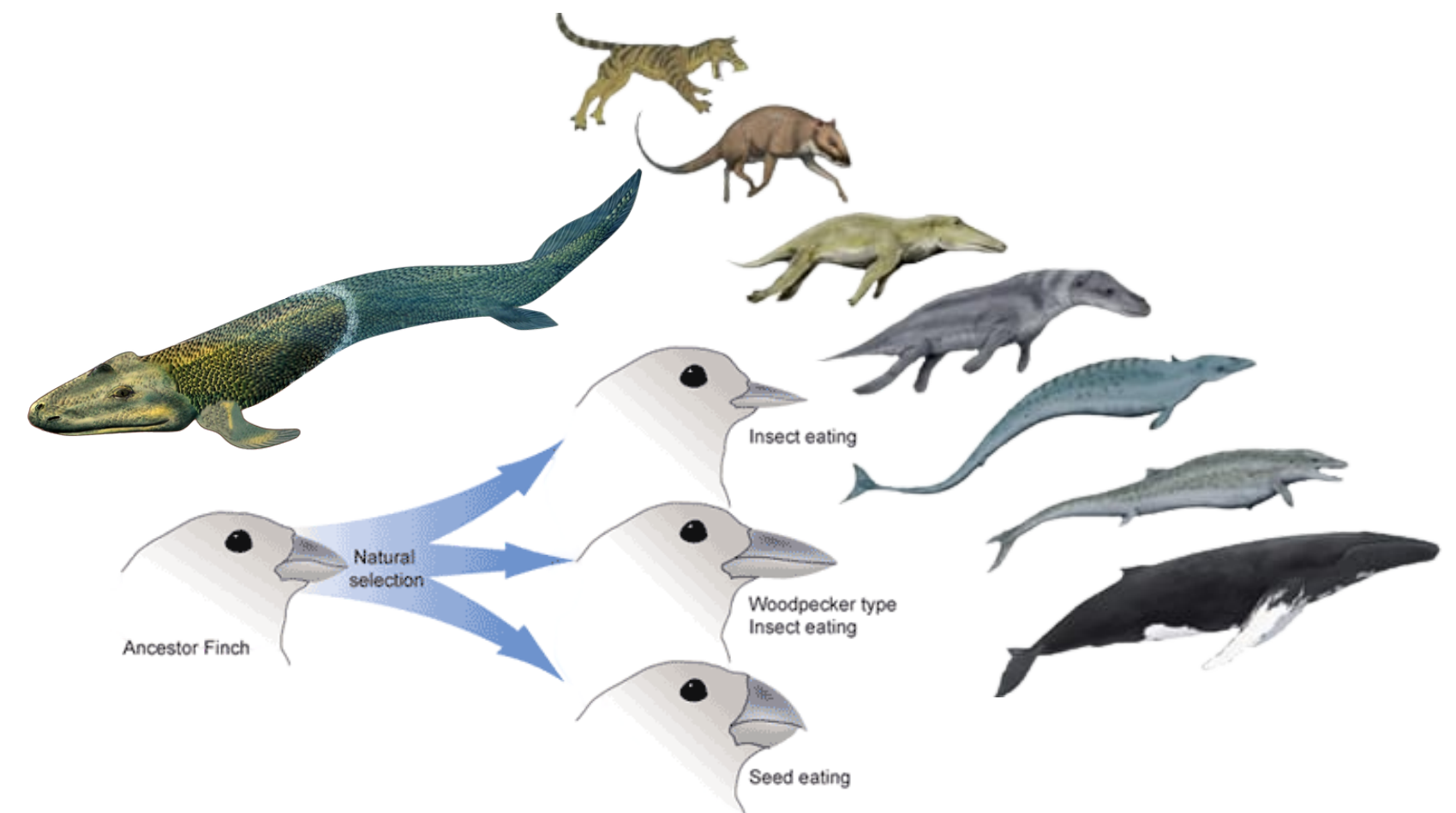
That -tree of life- we saw moments ago was not exactly built by time travelling.

All we have is sequences, genomes, proteins, sequencers, software and faith...

In order to *observe* how species evolved we need to measure distances between them.

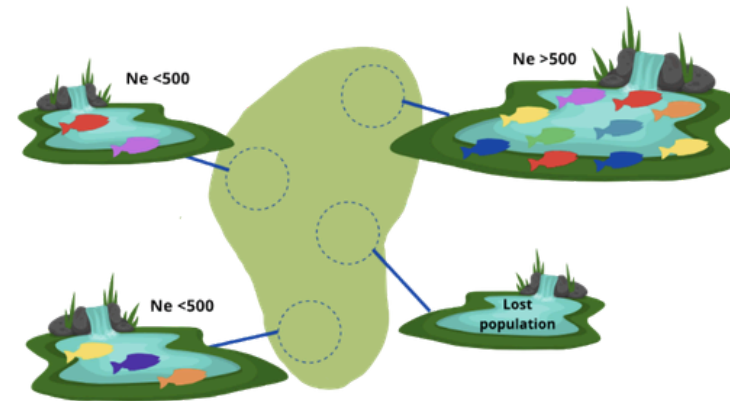
Genetic distance is a quantitative proxy for the accumulated evolutionary time between two lineages.

Broadly speaking: The longer the divergence time, the greater the accumulation of substitutions → greater distance.



Specific applications

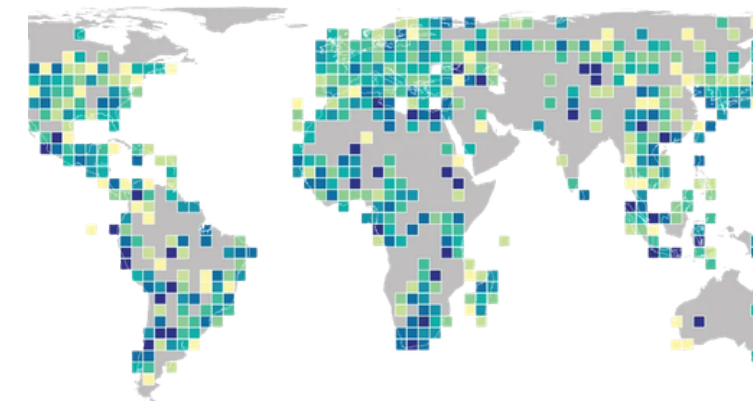
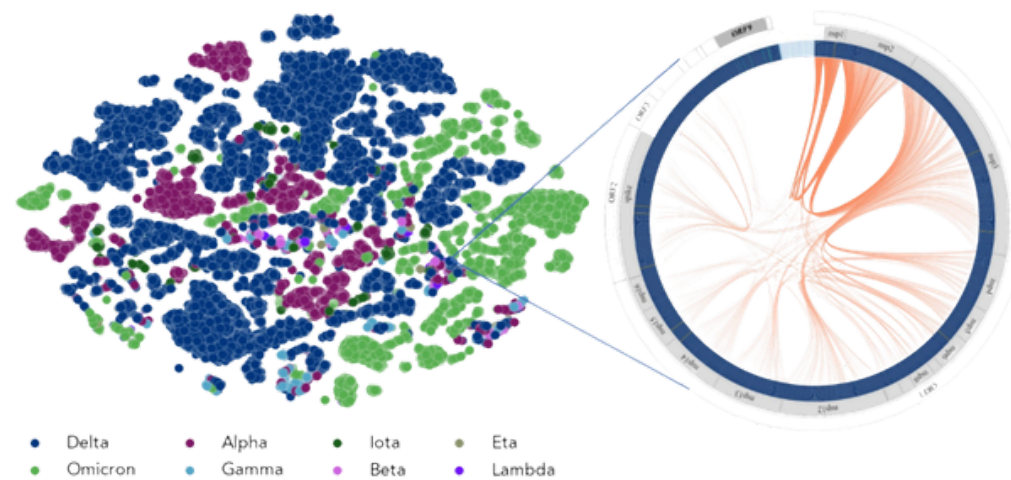
Molecular epidemiology: tracing the origin and rate of spread of pathogens. The distances between strains allow for the reconstruction of transmission chains.



Forensic medicine: identifying individuals or relating biological samples to populations of origin.



Biological conservation: measuring genetic diversity within and between populations to prioritize conservation efforts.



Agriculture and breeding: quantifying distances between varieties to design crosses that maximize genetic diversity.

Why isn't simply counting differences enough?

1

Counting distinct positions in an alignment gives the distance p (observed distance), which is the proportion of distinct sites:

$$p = \# \text{ differences} / \# \text{ total sites.}$$

2

The problem

When time is long, a site may have mutated twice and reverted to its original state (hidden substitution or homoplasy). This causes p to underestimate the actual distance.

3

Evolutionary models

That's why evolutionary models are needed to correct that underestimation.

METRICS

The hamming distance

- As we discussed briefly in the past, the hamming distance calculates the number of positions in which two sequences of **equal length differ**.
- The pillar of this approximation is the **XOR operator**.
- Assumption: each position is **independent**; it does not correct multiple substitutions or identify types of substitutions
- **Hamming distance does not consider time.**
Therefore, it will omit evolutionary bifurcations.

XOR (Exclusive OR) operation

Input A	Input B	XOR output
0	0	0
0	1	1
1	0	1
1	1	0

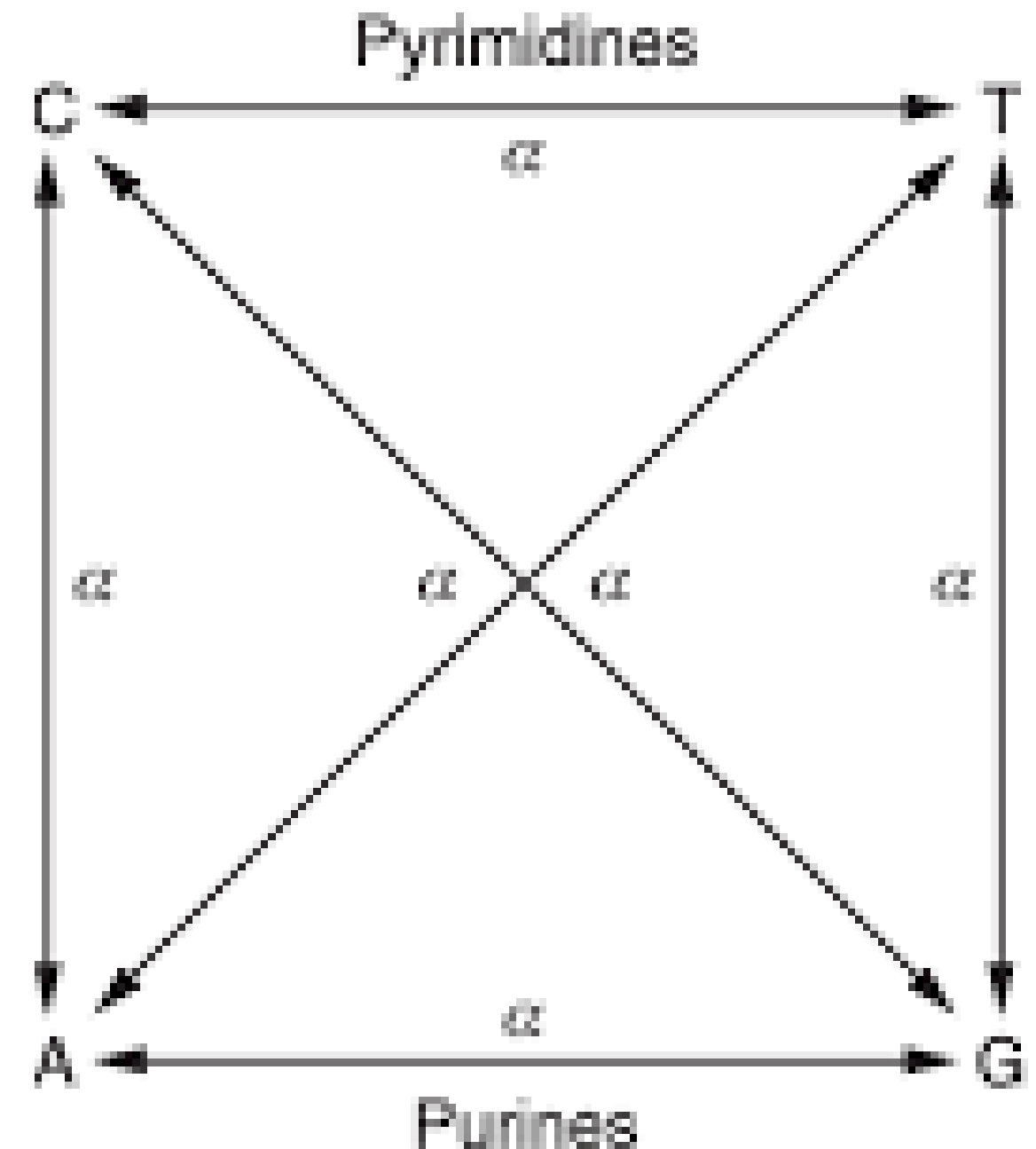
Hamming Distance = 2

Every time our XOR returns a 1 value, we have a dissimilarity and we add it.

$$d_H(A,B) = \sum [A_i \neq B_i]$$

The Jukes-Cantor distance

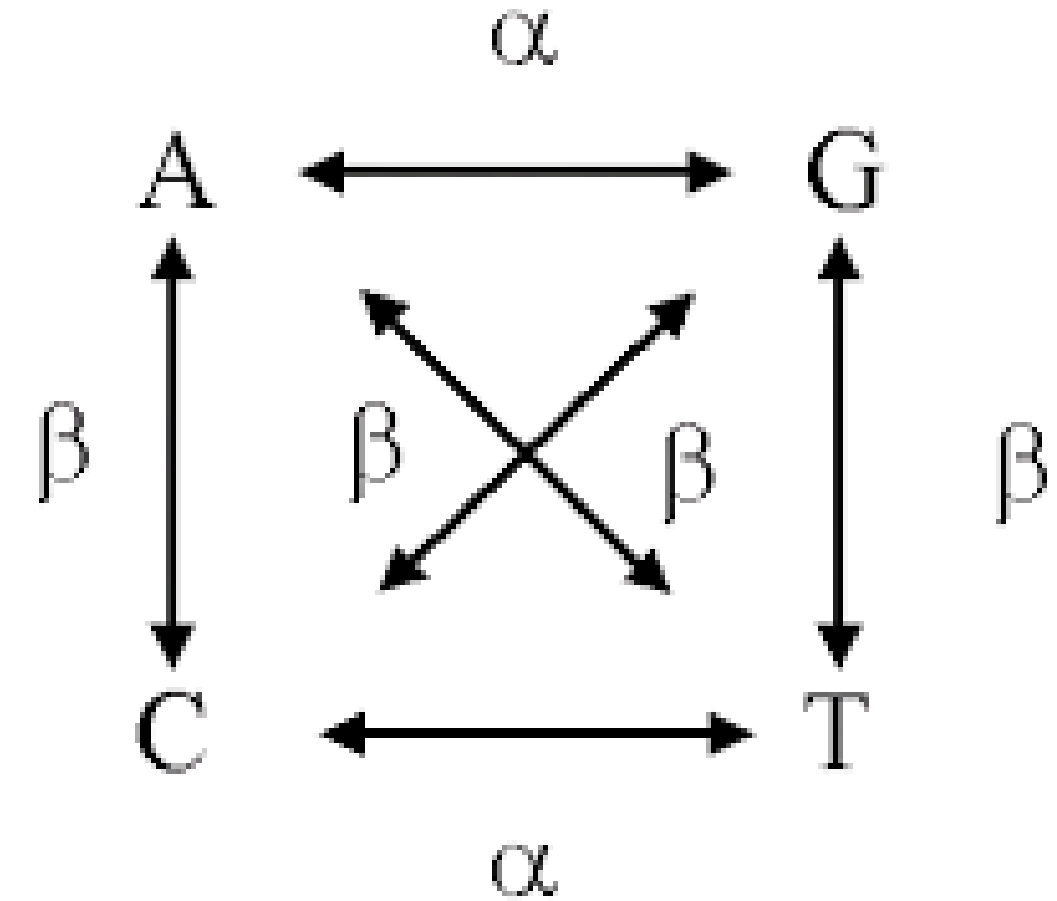
- The first substitution probabilistic model (1969)
- **Three principal assumptions:**
 - All four nucleotides have the **same frequency** (25% each)
 - All substitutions occur at the **same α rate**
 - Each site **evolves independently**
- It corrects the underestimation by estimating the actual number of substitutions per site.
- **THE LIMITATION!!** It does not distinguish between **transitions and transversions**. (In real genomes, transitions are more frequent than transversions)



$$d = -\left(\frac{3}{4}\right) \ln\left(1 - \frac{4p}{3}\right)$$

Kimura's 2-Parameter Model

- As Jukes-Cantor can't describe or distinguish *transversions and transitions*, Kimura introduced two different rate parameters:
 - α = rate of transitions ($A \leftrightarrow G, C \leftrightarrow T$)
 - β = rate of transversions ($A \leftrightarrow C, A \leftrightarrow T, G \leftrightarrow C, G \leftrightarrow T$)
- We are still assuming: equal nucleotide frequencies, independence between sites
- This models the real biological asymmetry between types of substitutions.



$$d = -\left(\frac{1}{2}\right) \ln(1 - 2P - Q) - \left(\frac{1}{4}\right) \ln(1 - 2Q)$$

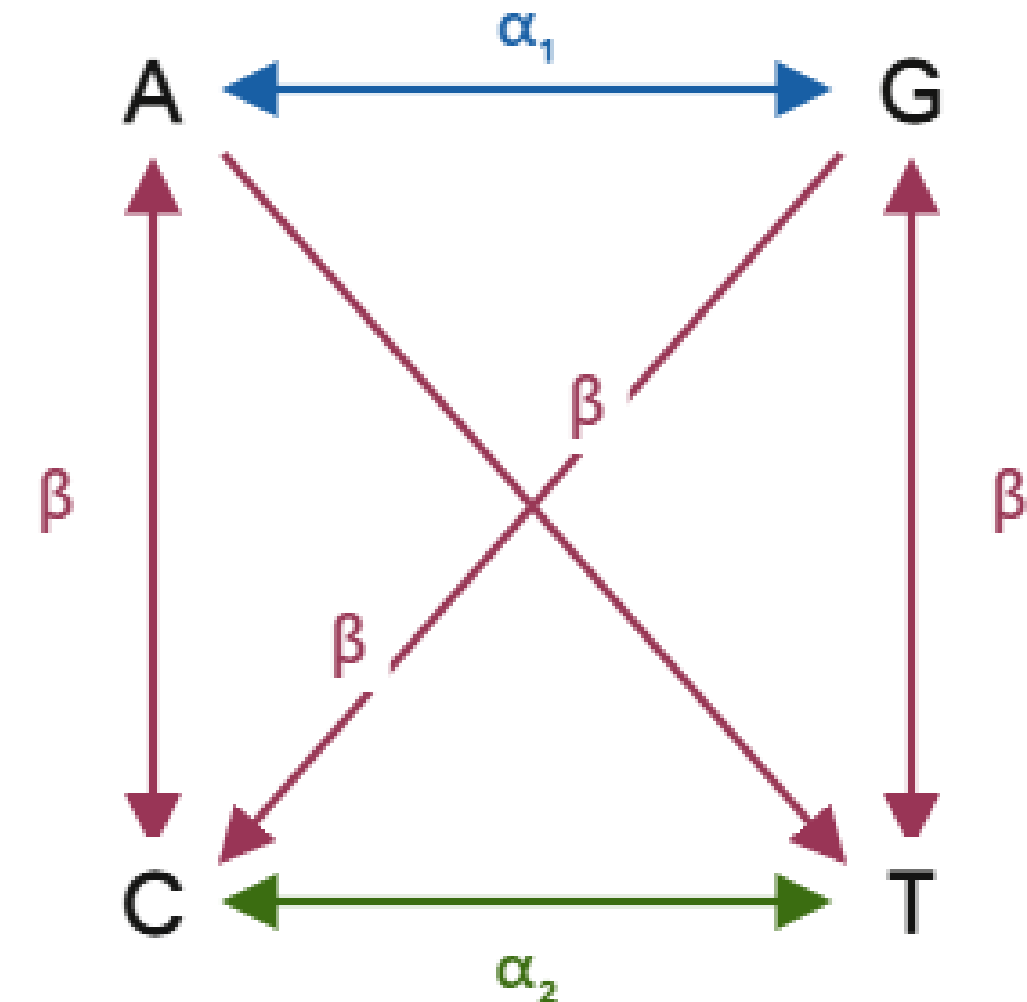
With

P = proportion of sites with transition

Q = proportion of sites with transversion.

Tamura-nei gamma distance

- Allows unequal base frequencies: $\pi_A, \pi_G, \pi_C, \pi_T$, thus, *captures the biological* reality that *$A \leftrightarrow G$ and $C \leftrightarrow T$ transitions can differ* from each other.
- Introduces three rate parameters:
 - α_1 = rate of transitions among purines ($A \leftrightarrow G$)
 - α_2 = rate of transitions among pyrimidines ($C \leftrightarrow T$)
 - β = rate of all transversions



$$d = -2\pi R \pi Y \left[\kappa_1 \frac{p_1}{(2\pi R \pi Y)} + \kappa_2 \frac{p_2}{(2\pi R \pi Y)} + q \right]$$

With

p_1 = proportion of purine transitions ($A \leftrightarrow G$)

p_2 = proportion of pyrimidine transitions ($C \leftrightarrow T$)

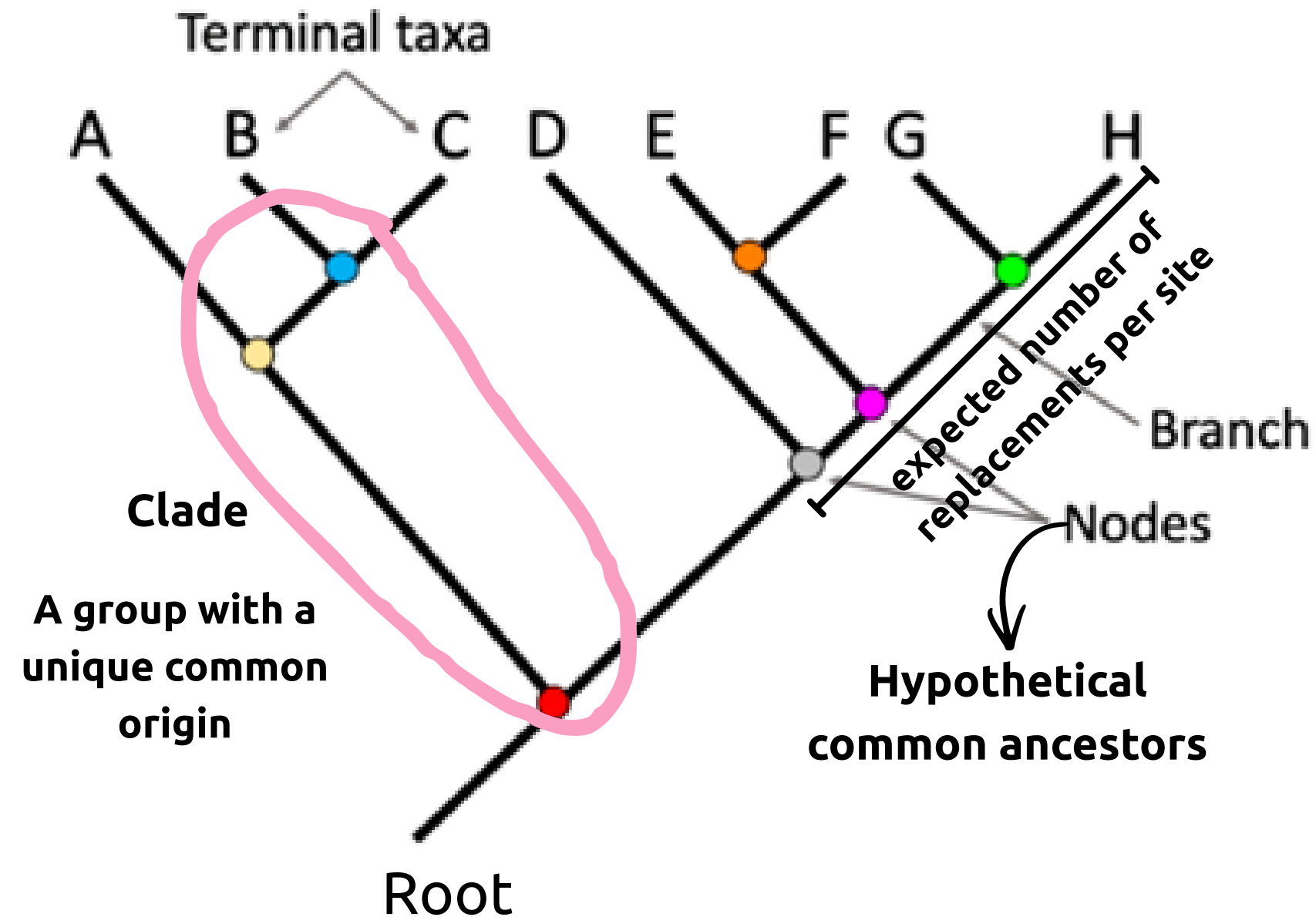
q = proportion of transversions

$\pi R = \pi A + \pi G \mid \pi Y = \pi C + \pi T$

WE SPEAK FOR THE TREES

What is a phylogenetic tree?

A phylogenetic tree is a map of who shares a common ancestor with whom, and when they diverged.



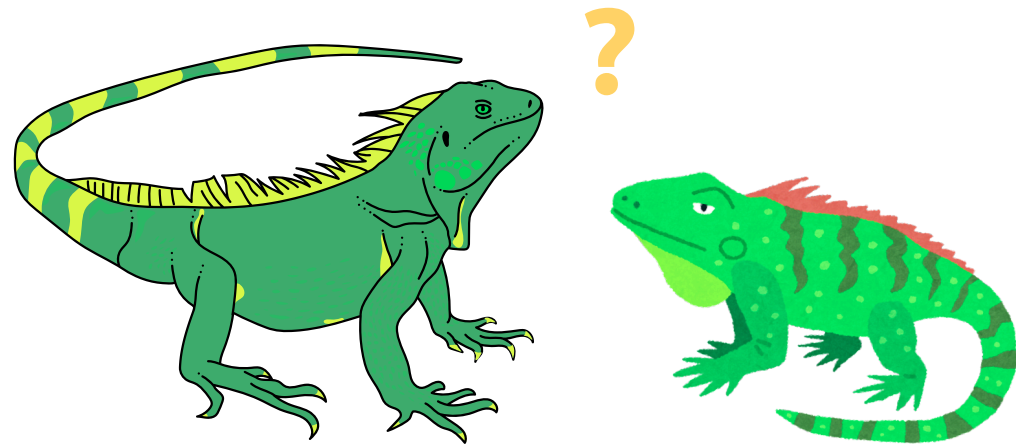
It's exactly like a family tree, but instead of people, it has species or lineages. Cousins share grandparents; sister species share a recent common ancestor.

- Which pairs of sequences diverged most recently (shortest branches between them)?
- Which lineages evolved most rapidly (longest branches)?
- How many divergence events occurred, and in what order?

CLUSTERING

What is clustering?

Remember our beloved iguanas? Well we are using them again



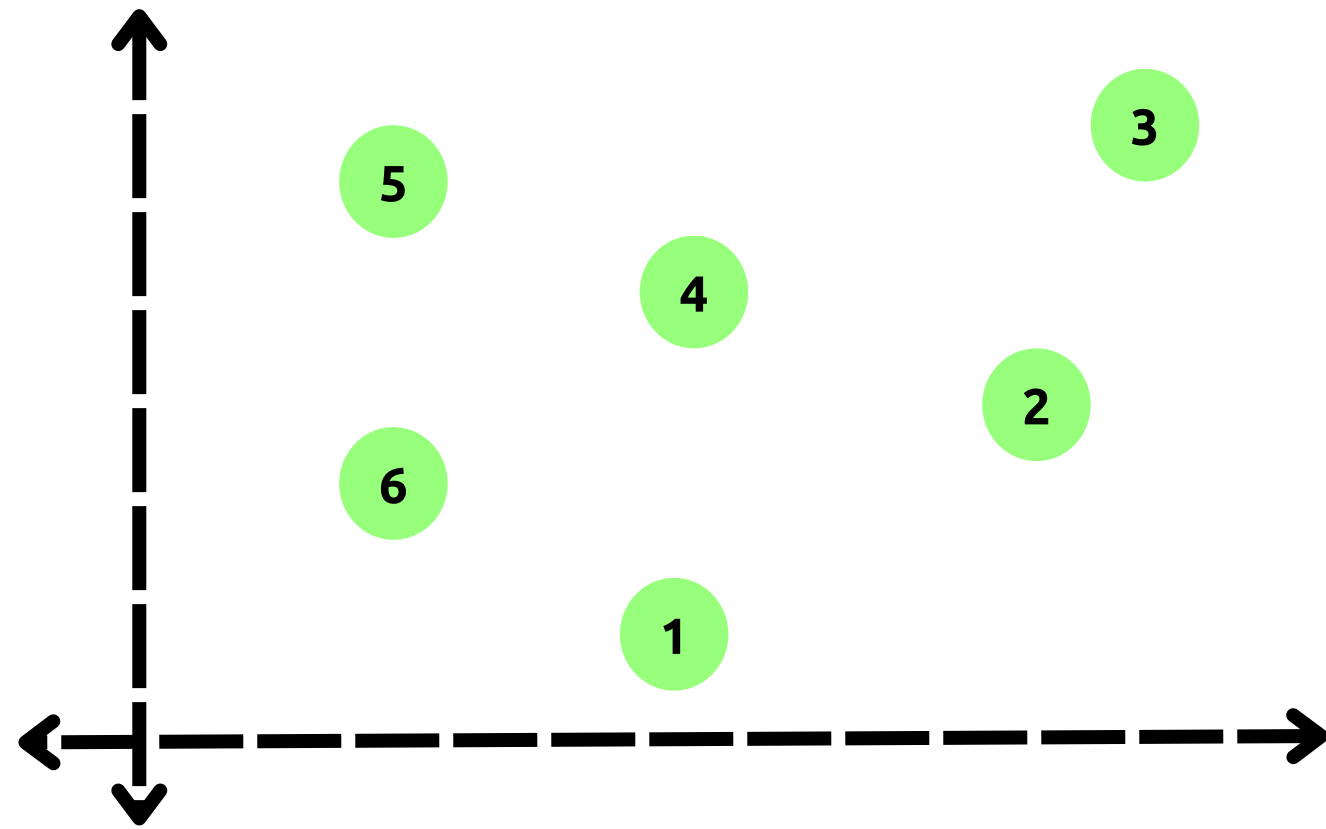
Having a set of distances between n sequences (an $n \times n$ distance matrix)

We can build a hierarchy of groups that reflects evolutionary relationships. (for example, how did two different species of iguanas evolved?)

Clustering means organizing a set of objects (sequences, expression profiles, etc.) into groups called "clusters". Our biological aim will be to **group biological functions or evolutionary relationships** maximizing Intragroup similarity and Intergroup dissimilarity.

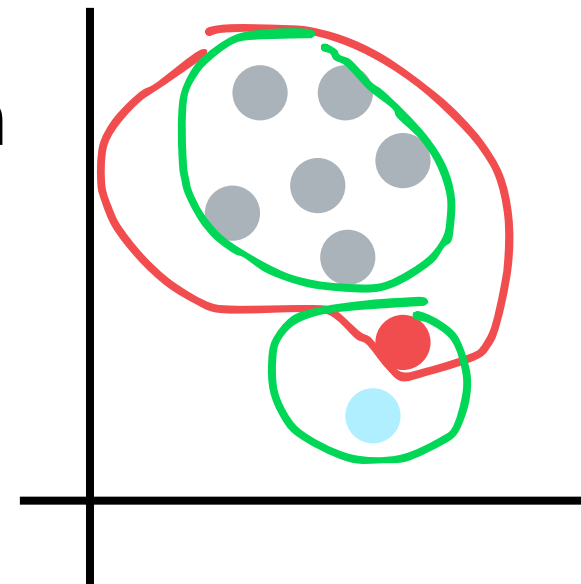
Agglomerative grouping

The "Bottom-Up" Approach



1. Calculate the initial **distance** matrix.
2. Identify the pair of elements with the smallest distance.
3. Join them into a new node (cluster).
4. Recalculate the distance matrix between the new cluster and the others.

It is assumed that each data point starts as an **individual cluster**. The algorithm proceeds iteratively, merging the two most similar clusters until they all are related



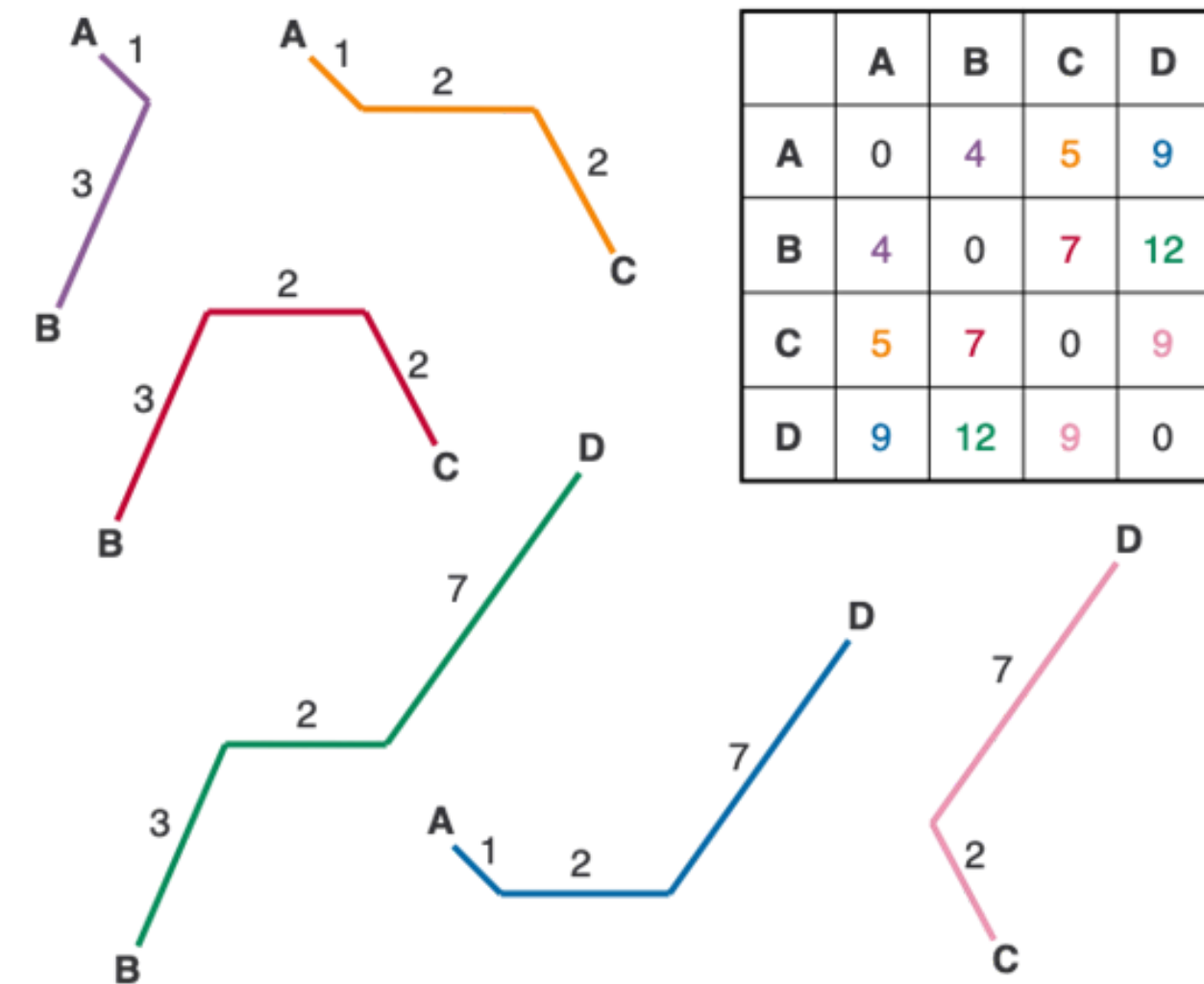
We mean related because spontaneous generation does not exist.

If you cluster the red dot with the biggest one, then the blue one would be unrelated. This cannot happen

Neighbor joining algorithm

Instead of simply joining the two closest clusters, NJ corrects the distances by the average branch length of each sequence, having as its main aim minimizing the total length of the tree (principle of minimum evolution).

Unlike other methods, NJ does not assume a constant molecular clock (equal mutation rate for everyone). This makes it much more accurate for real biological data.



$$d(ind\ i, ind\ j) = (n - 2)d_{ij} - R_i - R_j$$

$$R_i = \sum_k d(ind\ i, ind\ k)$$

Where R_i and R_j are the sums of rows and columns (dispersion parameters)

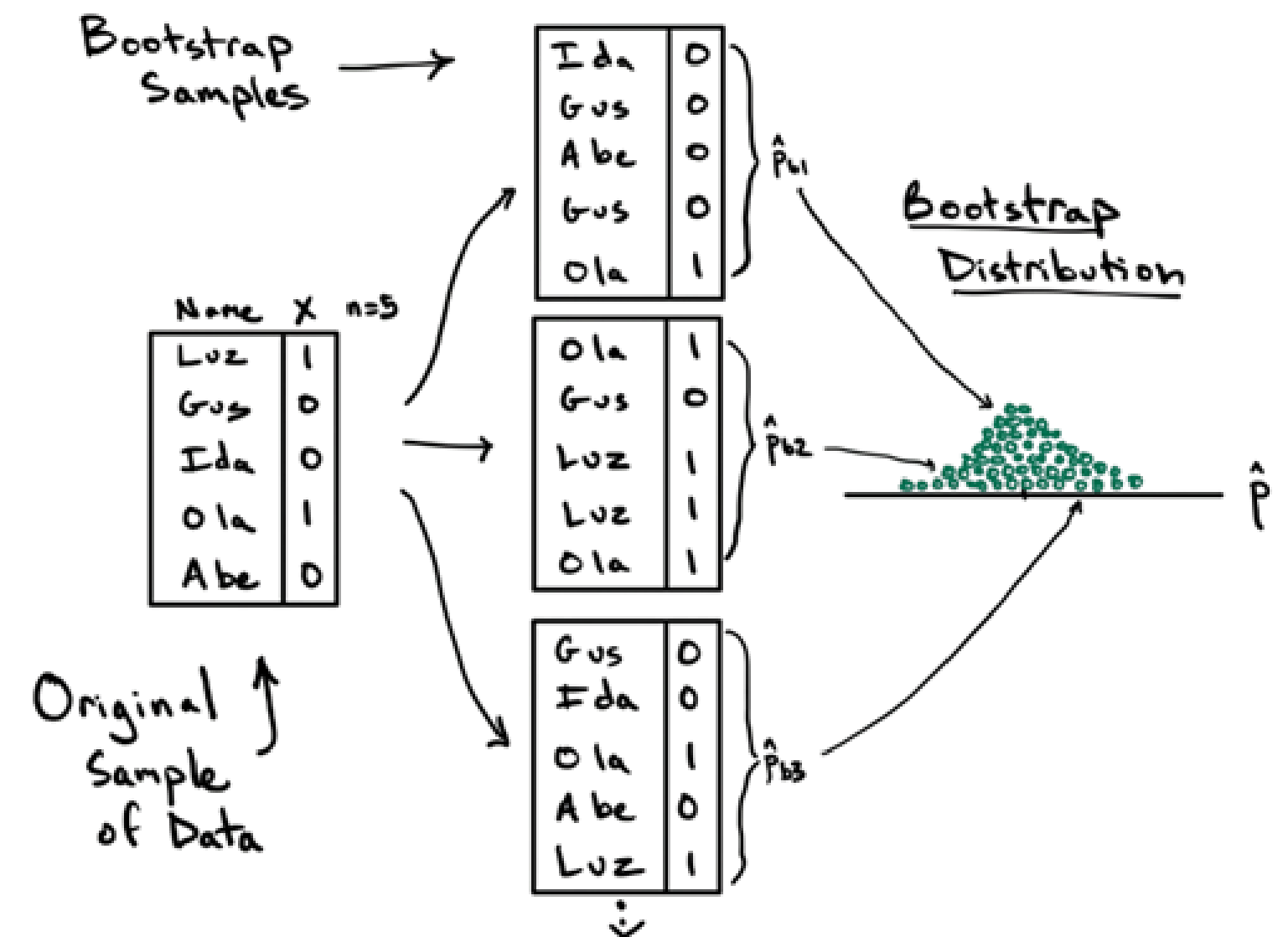
Bootstrapping / Re-sampling

If we had slightly different data, would we reach the same conclusion?

Imagine you have 1000 photos of the same landscape taken from slightly different angles. If the same tree appears in the center of 950 of them, you're pretty sure that tree is really there. If it only appears in 400, you have doubts.

For each node/cluster of the original tree you ask: in how many of the 1000 trees did this same clustering appear? That percentage is the bootstrap value of the node.

Bootstrap measures whether a node is reproducible under variations in the data. It does not guarantee that the tree is correct, only that it is consistent.



Thanks !!!!!